

Lab 3: Architectural Selection & Justification Document

System Name: Self-Service Coffee Kiosk System

Architecture Selection Statement: "We chose **Layered Architecture** for the Self-Service Coffee Kiosk System."

1. Architectural Choice & Scenario Justification

We selected a **3-Tier Layered Architecture** comprising Presentation, Business Logic, and Data Access Layers.

- **Hardware Integration & Single-Device Footprint:** The kiosk operates as a self-contained physical terminal attached to local hardware (Touchscreen, Thermal Printer, Card Reader). A Layered Architecture provides a clean design where drivers reside neatly in Business/Presentation layers without network latency.
- **Sequential Workflow & Low Complexity:** The coffee ordering lifecycle follows a predictable local flow (Select Drink/Size -> Authorize Payment -> Print Receipt -> Log Order), enforcing strict horizontal separation without microservice overhead.

2. Security Advantage & Performance Benefits

- **Security Advantage:** Isolates credit card authorization inside the **Payment Service Component** within the Business Layer. Direct UI input cannot access hardware registers or database credentials, meeting PCI-DSS guidelines.
- **Performance Benefit:** In-process function calls across layers guarantee sub-50ms UI response times during touchscreen navigation and instant local order state updates, eliminating network hop overhead entirely.

3. UML Component Diagram (5 Components, 4 Interfaces)

